

Rework: Waterfall Vs Agile – Comparison Table

Feature	Waterfall Model	Agile Model
Approach	Linear and sequential	Iterative and incremental
Process Flow	Moves step-by-step (Requirements → Design → Development → Testing → Deployment)	Continuous cycles (Plan → Develop → Test → Review → Repeat)
Flexibility	Rigid, changes are difficult once a phase is completed	Highly flexible; changes can be made anytime
Requirements	Must be clearly defined at the beginning	Can evolve throughout the project
Customer Involvement	Low; client reviews mainly at the end	High; client provides continuous feedback in every iteration
Delivery	Final product delivered after full development	Working increments delivered frequently (e.g., every Sprint)
Testing	Done after development is complete	Continuous testing within each iteration
Team Collaboration	Limited; roles are clearly defined and follow a structured hierarchy	High collaboration; cross-functional teams work together
Documentation	Extensive and mandatory at each stage	Minimal documentation; focuses on working software
Cost & Time	Higher cost if changes occur; timeline fixed	Cost-effective for changing requirements; timeline adaptive

Creating an Amazon Application Using the Waterfall Model

The **Waterfall Model** is a linear, sequential software development methodology.

Each phase must be completed **fully** before the next begins.

No going back.

REWORK: SCRUM & JIRA FRAMEWORK

Scrum

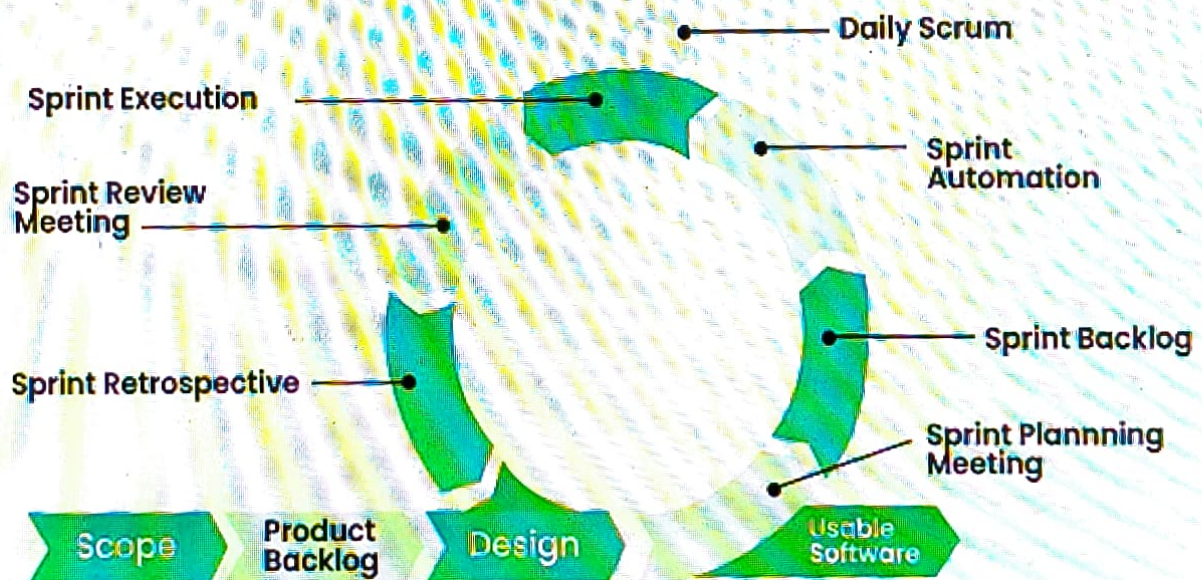
Scrum is an Agile framework used for developing complex products through collaboration, iterative progress, and continuous improvement. It organizes work into short, time-boxed cycles called *sprints* (usually 2–4 weeks), during which a cross-functional team delivers small, usable increments of the product. Scrum includes defined roles (Product Owner, Scrum Master, Development Team), key events (Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective), and artifacts (Product Backlog, Sprint Backlog, Increment) that help teams stay focused, respond quickly to change, and continuously improve both the product and the process.

Three Pillars of Scrum

1. **Transparency:** Everyone should clearly see the work and process.
2. **Inspection:** The team regularly checks progress and the product.
3. **Adaptation:** The team adjusts and improves based on what they learn.

Scrum Roles

1. **Product Owner:**
Represents the customer, decides what features are important, and manages the product backlog.
2. **Scrum Master:**
Supports the team, removes obstacles, and ensures Scrum rules are followed.
3. **Development Team:**
Builds and delivers a working piece of software at the end of each sprint.



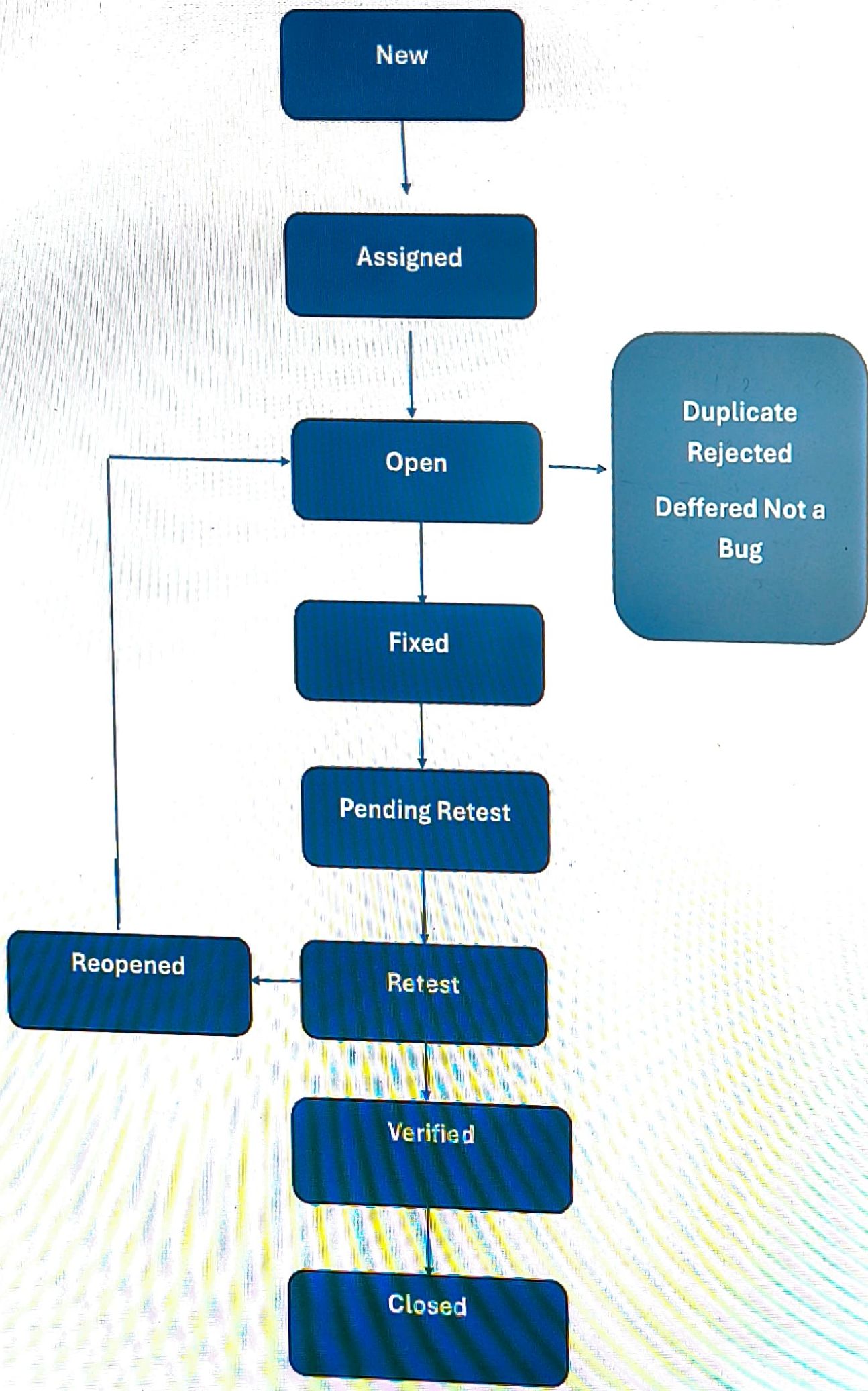
ACTIVITY – I

DEFEAT / BUG LIFE CYCLE

- A defect goes through different stages from the moment it is found until it is fixed and closed.
- Each stage shows the current status of the defect so the team knows what is happening with it.
- This cycle helps teams communicate clearly and work efficiently to track, assign, fix, and verify bugs.
- **Defect status**
- **Defect status** shows the current stage a bug is in during the defect life cycle.
- It helps the team understand **how far the bug has progressed** toward being fixed.
- The main goal is to **track, communicate, and manage defects clearly and efficiently** throughout the project.

Defect States Workflow

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ACTIVITY – II

IMPACT MATRIX

Severity / Priority	High Priority	Medium Priority	Low Priority
High Severity	• Complete site downtime • Payment gateway failure • Add to Cart not working for all products • Login system completely broken • Checkout failure for all users	• Login button not clickable • Category pages not opening • PDP (Product Detail Page) not loading important details • Search returning server errors • Registration submit button not responding	• Rare crash on some pages • Banner slider freezing occasionally • Product image zoom not working sometimes • Wishlist page loading slowly • Session timeout happening too early
Medium Severity	• “Place Order” button working inconsistently • Wrong prices shown on some product pages • Shipping charges not calculated • Tax not applied for some states • Cart quantity updates failing for some items	• Field validation issues in forms • Incorrect error messages • Pagination not working on category pages • Thumbnail images not switching on some PDPs • Sorting options not applying correctly	• Logical errors in UI (misaligned labels) • Incorrect breadcrumb link for one category • Product tags not showing on PDP • Recently viewed items not loading • Recommendations section incorrect
Low Severity	• Logo missing on some pages • Wrong favicon appearing • Spelling mistake in footer • Broken external link in contact page • Incorrect social media icons	• Field length restrictions not working as expected • Minor alignment issues in category list • Text overflow on Wishlist page • Slight delay in image loading • Hover effects not consistent	• Cosmetic issues (font mismatch) • Slight color variation in buttons • Extra white spacing in homepage sections • Minor CSS glitches on mobile view • Uncentered icons in header/footer